

# Documentation: *Olympics Analysis Dashboard*

## 1. Overview

This dashboard provides a comprehensive analysis of Olympic athletes, events, and performance metrics. It combines summary statistics with visualizations such as donut charts, treemaps, bar charts, and line graphs to highlight participation, gender distribution, medal counts, and long-term trends.

## 2. Dashboard Components

### Summary Metrics

- Total events: 723
- Total medals: 27,257
- Average BMI: 23
- Total athletes: 73,924
- Total games: 152,698 → These figures provide a snapshot of overall Olympic participation and performance.

### Gender Distribution (Donut Chart)

- Male: 73%
- Female: 27% → The Olympics historically show a strong male dominance in participation, though female representation has grown over time.

### Sport Share of Medals (Treemap)

- Largest shares: Athletics (2,791 medals), Swimming (2,618), Rowing (1,849).
- Other notable sports: Fencing, Ice Hockey, Hockey, Water Polo, Wrestling, Judo, Boxing. → Athletics and Swimming dominate medal distribution, reflecting their central role in Olympic competition.

### Total Athletes by Sport (Horizontal Bar Chart)

- Athletics: Over 11,000 athletes (highest).
- Other sports (e.g., Boxing, Canoeing, Football, Gymnastics) show smaller participation. → Athletics has the broadest athlete base, while niche sports have limited representation.

### Yearly Trend (Line Graph)

- Red line: Distinct athlete counts peaking at 5,394 in 2009, slightly declining to 5,067 in 2019.
- Black line: Steady growth from 217 in 1919 to 1,907 in 2019. ➡ Overall athlete participation has grown significantly, with fluctuations reflecting Olympic cycles.

#### 🌍 Medals by Country (Map Visualization)

- United States: 186 medals
- Canada: 29
- Japan: 1
- Russia: 0 ➡ The US dominates medal counts, while other countries show varied performance.

#### ⚖️ Age and BMI Correlation (Scatter Plot)

- Clustered around ages 20–40 with BMI near 23. ➡ Most Olympic athletes fall within a healthy BMI range, concentrated in early adulthood.

### 3. Key Insights

- Gender imbalance: Male athletes still dominate, though female participation is notable.
- Sport concentration: Athletics and Swimming account for the largest share of medals and athletes.
- Growth trend: Athlete participation has steadily increased over the decades.
- Country dominance: The United States consistently leads in medal counts.
- Athlete profile: Typical Olympians are aged 20–40 with BMI around 23.

### 4. Conclusion

The Olympics Analysis Dashboard highlights long-term growth in athlete participation, persistent gender disparities, and concentration of medals in core sports. It provides valuable insights into how demographics, sports specialization, and national performance shape Olympic history